
3AL Test 1 2022 Solutions

Time: 90 minutes

Available marks: 85

Maximum marks: 75

Instructions

1. Answer all questions in the lined answer booklets.
2. Start each question on a new page.
3. Make sure each answer is clearly labelled.
4. Show all work and reasoning.
5. This test consists of **4** questions and **2** numbered pages. Make sure your question paper is complete.

6. If you've read the instructions draw a smiley face on the front of your booklet :)

Good luck! I'll give you an extra mark if you do it, no joke

1. Let G be a group and let $H = Z(G)$.

- (a) Prove that if G/H (where $H = Z(G)$) is cyclic with generator gH , then every element of G can be written in the form $g^n h$ for $n \in \mathbb{Z}$ and $h \in H = Z(G)$.

Let G/H be cyclic with generator gH . Then for all $aH \in G/H$, we have $aH = (gH)^n$ for some $n \in \mathbb{N}^+$. Then $aH = g^n H$. Thus $a \in g^n H$ and so a can be written as $g^n h$ for some $h \in H = Z(G)$.

Notes

- G/H being cyclic does not imply that G is cyclic. (See Example 1.42)

- (b) Prove that if $G/Z(G)$ is cyclic, then G is abelian.

Suppose that $G/Z(G)$ is cyclic. Take $a, b \in G$. Then $a = g^n h$ and $b = g^m h'$ for

some $h, h' \in Z(G)$. Thus

$$ab = g^n h g^m h' = h g^n g^m h' = h g^{n+m} h' = h g^m g^n h' = g^m h h' g^n = g^m h' g^n h = ba.$$

Therefore G is abelian.

Notes

- 2 marks if student takes $n = m$

(c) Prove that if $|G : Z(G)| = 4$, then $G/Z(G) \cong K_4$.

Suppose $|G : Z(G)| = 4$. Then $G/Z(G)$ is isomorphic to K_4 or $\mathbb{Z}_4$. Suppose that $G/Z(G) \cong \mathbb{Z}_4$. So $G/Z(G)$ is cyclic and then G is abelian by (b). This means $G = Z(G)$ and so the factor group $G/Z(G)$ is $G/G = \{G\}$ with order 1. However, $|G/Z(G)| = |G : Z(G)| = 4$, so this is a contradiction. Therefore we must have $G/Z(G) \cong K_4$.

Notes

Isomorphic to K_4 or $\mathbb{Z}_4$ [✓⁵].

- G/H being cyclic does not imply that G is cyclic. (See Example 1.42)

Apply (b) [✓⁵].

Conclude that there is a contradiction [✓⁵].

[5 + 5 + 15 = 25 marks]

2. (a) State the First Isomorphism Theorem (any version).

First Isomorphism Theorem. Let $\varphi : G \rightarrow G'$ be a group homomorphism. Then $\ker \varphi \triangleleft G$ and

$$G/\ker \varphi \cong \text{im } \varphi.$$

Notes

- Could also use version where φ is epimorphism.
- Don't need to explicitly state that $\ker \varphi \triangleleft G$.
- No marks if student says "there exists/we can construct a homomorphism $\varphi : G \rightarrow G''$ ".

(b) Let $\varphi : G \rightarrow G'$ be an epimorphism. Suppose that B is a normal subgroup of G' and let $A = \{g \in G : \varphi(g) \in B\}$. Construct an epimorphism $\tilde{\varphi} : G \rightarrow G'/B$ and use the First Isomorphism Theorem to show $A \triangleleft G$.

Let $\tilde{\varphi} : G \rightarrow G'/B$ be defined by $\tilde{\varphi}(g) = \varphi(g)B$. Then $\tilde{\varphi}$ is a homomorphism since, for $g_1, g_2 \in G$, we have

$$\tilde{\varphi}(g_1 g_2) = \varphi(g_1 g_2)B = \varphi(g_1)\varphi(g_2)B = \varphi(g_1)B\varphi(g_2)B = \tilde{\varphi}(g_1)\tilde{\varphi}(g_2).$$

All elements of G'/B can be written as $g'B$ for some $g' \in G'$. Since φ is an epimorphism, there exists $g \in G$ such that $\varphi(g) = g'$. Thus $\tilde{\varphi}(g) = \varphi(g)B = g'B$. So $\tilde{\varphi}$ is an epimorphism.

Note that $g \in G$ is an element of $\ker \tilde{\varphi}$ if and only if $\tilde{\varphi}(g) = B$. That is $\varphi(g)B = B$. Equivalently, $\varphi(g) \in B$. That is, $g \in A$. Thus $\ker \tilde{\varphi} = A$.

By the First Isomorphism Theorem, $A \triangleleft G$.

Notes

Define map & show homomorphism [✓⁵].

- 2 marks for correct map.
- Not necessary to prove well-defined.
- If the map doesn't work for the question, no marks for this or the following parts.

Epimorphism [✓⁵].

Kernel & $A \triangleleft G$ [✓⁵].

[5 + 15 = 20 marks]

3. (a) State the Third Isomorphism Theorem.

Third Isomorphism Theorem. *Let H and K be normal subgroups of G with $H \subseteq K$. Then $K/H \triangleleft G/H$ and*

$$(G/H)/(K/H) \cong G/K.$$

Notes

Third Isomorphism Theorem [✓⁵].

- Not necessary to state $K/H \triangleleft G/H$, but everything else is necessary.
- Lose 1 mark for not saying H is normal, 1 mark for not saying K is normal, 1 mark for not saying $H \subseteq K$, 2 marks for not getting the right isomorphism.
- Can use letters other than G , H and K .

(b) Let a and b be positive integers such that $a \mid b$. Determine the order of the factor group $a\mathbb{Z}/b\mathbb{Z}$.

Note that since $a \mid b$, we have $b\mathbb{Z} \subseteq a\mathbb{Z} \subseteq \mathbb{Z}$. Moreover, since $\mathbb{Z}$ is abelian, these subgroups are normal. By the Third Isomorphism Theorem,

$$(\mathbb{Z}/b\mathbb{Z})/(a\mathbb{Z}/b\mathbb{Z}) \cong \mathbb{Z}/a\mathbb{Z}.$$

However, $\mathbb{Z}/a\mathbb{Z} \cong \mathbb{Z}_a$ and so $|\mathbb{Z}/a\mathbb{Z}| = |\mathbb{Z}_a| = a$. Similarly, $|\mathbb{Z}/b\mathbb{Z}| = b$. By Lagrange's Theorem,

$$|(\mathbb{Z}/b\mathbb{Z})/(a\mathbb{Z}/b\mathbb{Z})| = \frac{|\mathbb{Z}/b\mathbb{Z}|}{|a\mathbb{Z}/b\mathbb{Z}|} = \frac{b}{|a\mathbb{Z}/b\mathbb{Z}|}.$$

However, due to the Third Isomorphism Theorem, this must equal $|\mathbb{Z}/a\mathbb{Z}| = a$. Thus

$$a = \frac{b}{|a\mathbb{Z}/b\mathbb{Z}|} \implies |a\mathbb{Z}/b\mathbb{Z}| = \frac{b}{a}.$$

Notes

Third Isomorphism Theorem [✓⁵].

- For full marks, have the correct G , H and K plugged correctly into the Third Isomorphism Theorem.
- Lose one mark if they write $=$ instead of $\cong$.

Order of $\mathbb{Z}/a\mathbb{Z}$ [✓⁵].

- Use the fact that $\mathbb{Z}/a\mathbb{Z} \cong \mathbb{Z}_a$ to say that $|\mathbb{Z}/a\mathbb{Z}| = a$. Should actually mention this fact.
- No marks if the student writes $|\mathbb{Z}/a\mathbb{Z}| = |\mathbb{Z}|/|a\mathbb{Z}| = |\mathbb{Z}|/a|\mathbb{Z}| = 1/a$ or something similar. Lagrange's Theorem only works for *finite* groups.

Order of $a\mathbb{Z}/b\mathbb{Z}$ [✓⁵].

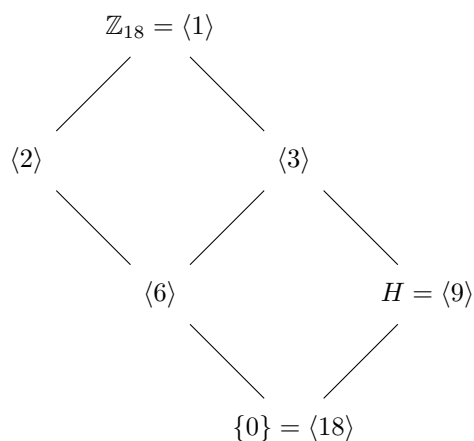
- Can only earn marks if previous parts were also correct, i.e. can't randomly come up with right answer.

[5 + 15 = 20 marks]

4. Consider the group $G = \mathbb{Z}_{18}$ with subgroup $H = \langle [9] \rangle$.

(a) Determine all subgroups of $\mathbb{Z}_{18}$ and draw the lattice diagram.

By the Fundamental Theorem of Finite Cyclic Groups, the subgroups of $\mathbb{Z}_{18}$ correspond to the divisors of 18. So, 1, 2, 3, 6, 9 and 18. The subgroups of $\mathbb{Z}_{18}$ are the unique groups of these orders. Namely, $\langle 18 \rangle$, $\langle 9 \rangle$, $\langle 6 \rangle$, $\langle 3 \rangle$, $\langle 2 \rangle$ and $\langle 1 \rangle$. The lattice diagram is



Notes

Divisors & subgroups [✓⁵].

- It's fine to drop the equivalence class notation.
- If a student does not show how they obtained the subgroups from the divisors, then they only get 2 marks. Provided the subgroups are correct. This also holds if they do not write the subgroups out explicitly but the subgroups in the lattice diagram are correct.
- Do not need to explicitly mention the Fundamental Theorem of Finite Cyclic Groups.
- Do not need to show the elements of each of the subgroups.
- No marks if the subgroups are written as $\mathbb{Z}_2$, $\mathbb{Z}_3$, etc. (Because those aren't subgroups of $\mathbb{Z}_{18}$. See Hand-in 2 Q1a.)
- It's fine if the subgroups are written as $2\mathbb{Z}_{18}$, $3\mathbb{Z}_{18}$, etc.
- Fine to write $\mathbb{Z}_{18}$ as $\langle 1 \rangle$ and $\{0\}$ as $\langle 18 \rangle$ or $\langle 0 \rangle$. Or any other funky correct thing, I guess.

Lattice diagram [✓⁵].

- Beauty matters not, so long as all the lines and groups are in the right place.
- No marks if the diagram is upside down.

- Still no marks if subgroups are being referred to as $\mathbb{Z}_2$, $\mathbb{Z}_3$, etc.
- Otherwise, no partial marks. Diagram needs to be right to earn full 5 marks.

(b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{18}/H$.

$$\begin{array}{c}
 \mathbb{Z}_{18}/H \\
 | \\
 \langle 3 \rangle / H \\
 | \\
 H/H = \{H\}
 \end{array}$$

Notes

Lattice diagram [✓¹⁰].

- Beauty matters not, so long as all the lines and groups are in the right place.
- No marks if the diagram is upside down.
- No marks if the groups are not factor groups.
- Fine to write “/ H ” or “/ $\langle 9 \rangle$ ”.
- Still no marks if subgroups are being referred to as $\mathbb{Z}_2$, $\mathbb{Z}_3$, etc.
- Otherwise, no partial marks. Diagram needs to be right to earn full 10 marks.
- If students include a subgroup that does not belong to the above diagram, 0 marks.
- Fine to write H/H as $\{H\}$.

[10 + 10 = 20 marks]